

## **BSCHE Graduate Written Examination – Fall 2025**

Written examination conducted with graduating seniors in Chemical Engineering

Name: Kyle Hankosky

---

### **Question 1**

#### **Career plans. (Scoring rubric was designed to assess potential for life-long learning.)**

Where do you see yourself in 5-10 years, and what steps will you take to get there?

As a process engineer in pharmaceutical manufacturing, I interned and now have a full time offer for a biopharmaceutical company. And from my experiences there I know it isn't the place I want to spend my whole career but at least the next 4-6 years to build experiences both in project management and general engineering skills

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Currently I have an offer for a biopharmaceutical company called Pfanstiehl in Waukegan IL, where I will be a process engineer this is exactly where I wanted to be as an engineer. Throughout the 4 years at uic hearing others experiences and my own research I knew that I wanted to end up in manufacturing. R&D has never really interested me. And I knew that I wanted to end up either in pharmaceuticals or specialty chemicals production. Sadly this company doesn't have any kind of mentorship or rotational programs which might inhibit my career growth especially as there isn't any career growth at this company I'm effectively 3 promotions away from the CEO. I have done some research into getting my PE, but after understanding what this allows me to do as a chemical engineer I really don't think it is necessary.

Does your prospective employer sponsor master's degree or MBA? Under which terms?

They do not, with how lucrative the biopharma industry is to work in as a process engineer for this company so far no one has requested this opportunity.

## **Question 2**

### **Professional ethics and ethical ramifications of technology.**

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

The FDA's new regulations on endotoxin concentrations for cell and gene therapies. The positives is more stringent regulation on injectable biologics making these therapies safer for patients. The negatives this increased regulation on contaminants increases the difficulty of manufacturing these components with greatly increased cost as they move the goal posts on the industry. Also there are very few companies in the world that can produce excipients at these levels of purity which raises monopoly concerns for the industry

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

At my current job here at uic one of my colleagues was consistently not performing her job responsibilities, consistently scheduling class project meetings during work hours and finding any excuse of why she could not help us complete our duties along with our job requires a "Strong background in organic and inorganic chemistry" to complete our duties and she lacked these and misrepresented her knowledge on the subject during her interview process. Along with this she would consistently log hours she did not work or leave work early and still log the hours she was not there. The culmination of these actions for 6 months was her termination from the department due to her rampant wage theft.

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

It really depends on the specific dilemma, if we are talking about actions that would cause harm such as operation of a unit in a manner that will cause harm absolutely I am turning to higher powers to get the situation under control. But for instance if a friend was cheating on their s/o it is much more complicated. I feel that there is no systematic methodology that can resolve this it would be my gut feeling with my personal experience with the person that would be the deciding factor in my reaction.

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it? Yes, provision 6 "Maintain and improve professional competence."

Which courses at UIC have contributed most to your views on professional ethics? How? CHE 210, this is the only course where professional engineering ethics were at all mentioned so far 396 had one lecture on safety but not about professional competence as an engineer it was assumed. Honestly the sum of the professional ethics I have learned at uic can be summed up in one quote “you have a professional responsibility to not blow up the plant” and sadly that’s it. Especially after talking to my peers at my internship their universities had at minimum an 8 week course taken senior year on engineering ethics and I think this is a serious deficiency in our curriculum here.

---

**Question 3**

**Working in multidisciplinary teams**

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

---

**Question 4**

**Numerical methods and computational proficiency (Excel, MATLAB or VBA): choose one problem.**

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} \left[ x_{\text{in}}(t) - x(t) \right] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants  $\tau = 3.4$ ,  $k = 1.2$ ,  $\varepsilon = 0.1$  and  $\omega = 5$ . Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

---

I chose part B and to solve it in Fortran, this is the language I have the most proficiency in as MATLAB is very poorly taught in CS109 and vba is very clunky to use and program in.

```
program gaussian_integration
  implicit none
  real(8) :: nodes(10), weights(10)
  real(8) :: a, b, integral, x, jacobian
  integer :: i

  !limits
  a = 1.0d0
  b = 7.0d0

  !nodes
  nodes(1) = -0.9739065285171717d0
  nodes(2) = -0.8650633666889102d0
  nodes(3) = -0.6794095682990244d0
  nodes(4) = -0.4333953941292472d0
  nodes(5) = -0.1488391745338768d0
  nodes(6) = 0.1488391745338768d0
  nodes(7) = 0.4333953941292472d0
  nodes(8) = 0.6794095682990244d0
  nodes(9) = 0.8650633666889102d0
  nodes(10) = 0.9739065285171717d0

  !weights
  weights(1) = 0.0666713443086881d0
  weights(2) = 0.1494513491505306d0
  weights(3) = 0.2190863625159820d0
  weights(4) = 0.2692567250960377d0
  weights(5) = 0.2955242247147529d0
  weights(6) = 0.2955242247147529d0
  weights(7) = 0.2692567250960377d0
  weights(8) = 0.2190863625159820d0
  weights(9) = 0.1494513491505306d0
  weights(10) = 0.0666713443086881d0

  jacobian = (b - a) / 2.0d0

  integral = 0.0d0
  do i = 1, 10
    x = a + (nodes(i) + 1.0d0) * jacobian
    integral = integral + weights(i) * 1/(1+cos(x)**3+log(x))
  end do

  integral = integral * jacobian

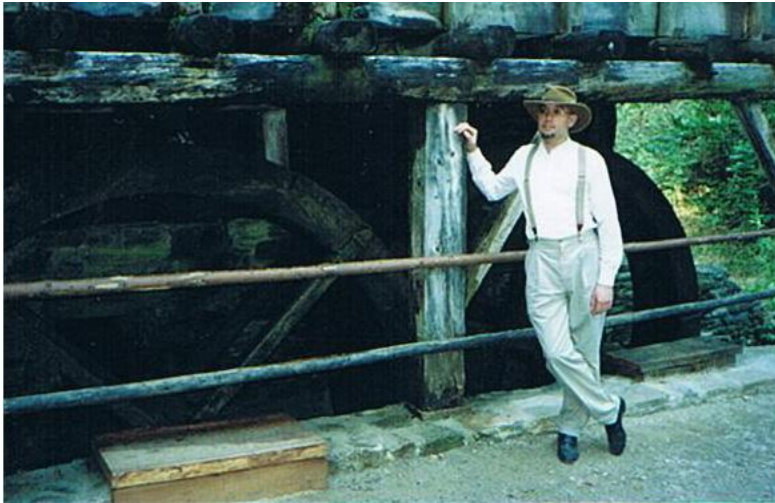
  print *, "Result:", integral
end program gaussian_integration

Result: 3.1421381005962097
```

### **Question 5**

**Engineering Methodology Proficiency: Engineering estimate.**

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?



---

Starting off with estimating the diameter of the water wheel, assuming that the average height a man in the USA is 5'9" which we will assume this man is. And further assuming that this man's head + neck is ~10" we will assume that the radius of the water wheel is ~4'10", there are 6 buckets per 45 degrees of wheel, that the width of the buckets are 1' and water holding capacity of each bucket is 6".

So

Total Buckets = 48

Height of falling water into bucket = 10

Circumference = 30.37 ft

Depth of bucket = 0.5 ft

Volume of bucket =  $0.316 \text{ ft}^3$

Volume of water per full bucket = 2.4 gal

RPS = 0.067s<sup>-1</sup>

Volume of water per revolution =  $48 * 0.316 = 15.2 \text{ ft}^3$

Volume flow =  $15.2 * (1/15) = 1.01 \text{ ft}^3/\text{s}$

Volume flow =  $1.01 * 7.48 * 60 = 453.3 \text{ GPM}$

And using the following equation for shaft HP with an assumed efficiency of 0.7 for this wheel

$\text{HP} = 0.113 * Q * H * \text{efficiency} = 0.8 \text{ HP}$

